

From Neural Trajectories to Cross-Scale Approximation: The Mathematics Program of Self-Aware Networks

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Abstract

A neural event is not received by an empty system. It arrives at a learned receiver, along a delayed route, inside an ongoing trajectory, and its consequence can be transformed again before becoming available at another biological scale. This creates a mathematical problem that cannot be solved by assigning one number to a spike or one coordinate system to the brain. The model must keep heterogeneous measurements typed, represent receiver dependence, preserve temporal order, translate among local coordinate systems, compose processes through declared interfaces, and distinguish a decoded endpoint from the mechanism that produced it. This paper organizes nineteen mathematical families developed across the Self Aware Networks (SAN) program into one executable, modular framework. Twenty-four numbered equations define state trajectories, typed receiver maps, receiver-relative event differences, local approximation, history, coordinate transport, compositional process maps, invariance and equivariance, partial-view reconstruction, cross-scale dynamics, model selection, endpoint equivalence, intervention separation, and latent-coordinate ambiguity. The paper's distinct contribution is a deterministic synthetic benchmark that asks whether model selection can recover four minimum adequate structures rather than rewarding the most elaborate model by default. Across 30 fixed seeds, the selection rule recovered the intended instantaneous, receiver-dependent, history-dependent, and transported-composition model in 30 of 30 runs per regime. Median held-out root-mean-square error in the cross-scale regime fell from 1.178 for the instantaneous model and 0.319 for the untransported history model to 0.131 for the transported-composition model. In a second test, two routes were constructed to have nearly identical observed endpoints. Source-plus-endpoint features classified route identity at 50.0%, while typed intermediate states classified it at 100%; blocking an intermediate transition exposed a mean route gap of 2.213 despite a normal endpoint gap of 0.020. These results validate the benchmark logic, not a biological implementation. The empirical program therefore requires preregistered neural variables, held-out transformations, selective route perturbation, and receiver-specific rescue before any part of the framework is treated as a neural mechanism. Keywords: Self Aware Networks; neural dynamics; receiver dependence; cross-scale modeling; compositionality; state space; sheaf transport; Petri nets; tomography; model selection

1. The ordinary problem comes before the mathematics

You touch a cup, recognize its shape, adjust your grip, and avoid dropping it. The useful event is not one unchanging packet that begins in the skin and ends in the hand. Receptor deformation changes ionic currents. Those currents meet cells whose dendrites, conductances, inhibition, and recent history differ. Population activity develops over time. Visual, tactile, proprioceptive, and motor populations use different measurements. Their outputs converge on action, and the action returns new sensory evidence.

The mathematical problem is therefore not simply to decode a label from neural activity. It is to explain how a relation remains useful while it is repeatedly transformed. A successful account must answer six questions in order:

1. What state is changing through time?
2. Which receiver transforms an arrival, and what learned state does that receiver bring?
3. Which difference relative to that receiver predicts a consequence?
4. How are histories and delays represented?

5. How are locally different coordinates transported and composed?
6. Which observation, perturbation, and rescue would distinguish the proposed route from a simpler alternative?

Only after that problem is visible do the SAN terms become useful. Expert Data Structure (EDS) names a proposed learned receive-transform-project configuration. Phase Wave Differential (PWD) names a proposed typed, receiver-relative event difference. Neural Array Projection Oscillation Tomography (NAPOT) names a repeated partial-view reconstruction hypothesis. Non-linear Differential Continuous Approximation (NDCA) names the remaining cross-scale problem: how action-relevant distinctions remain usable through nonlinear, delayed, partial, and continuously changing transformations. The acronyms summarize operations that have already been explained; they do not substitute for the measurements.

The central paper-level question is narrower than a complete mathematical theory of consciousness:

Can a typed, delay-aware, receiver-relative composition model identify which mathematical structure is needed for a declared prediction, while rejecting unnecessary structure and preserving differences that an endpoint-only analysis erases?

The answer is tested here on synthetic systems whose generating structure is known. The benchmark is deliberately capable of returning a simpler model. It is not designed to make the full framework win automatically.

2. A source-bounded development from 2012 to 2026

The mathematics program did not appear as one finished formalism. It developed through a series of questions and increasingly specific operators. Notes carrying 2012 author dates asked whether neurons, axons, firings, synapses, vectors, matrices, directed graphs, topologies, categories, and differential equations could describe the flow of mental patterns [1-4]. Those notes were publicly fixed in Git in 2022, so the embedded author dates and public fixation dates remain separate. They establish an early research direction, not a completed category, neural vector code, or differential model.

A 2014-dated note compared active dendritic memory with a self-activating content-addressable structure and used the phrase Expert Data Structure [5]. A public 2017 Neural Lace Podcast then posed the network-protocol question: if the brain is a network, when does communication require feedback and when can it proceed without an acknowledgment? The discussion used TCP and UDP as engineering contrasts and proposed Alternating Transmission Protocol as a biological research analogy [6]. The analogy raised the right problem but did not make neural tissue an Internet stack.

Public Git sources from June and July 2022 made the program more specific. A June 8 note proposed tracking states and transitions at nested cellular places through Petri-net and category-theory language [7]. July notes joined successive phase-state changes to Taylor-series approximation, tensors, inhibition, route selection, reaction-diffusion, and multidimensional trajectories [8-10]. The recoverable contribution was not the invention of Taylor series, phase space, or Petri nets. It was the attempt to join learned receivers, ordered physical differences, nested biological places, and downstream consequences in one neural modeling program.

The first currently verified named public NDCA route was Git-fixed on January 5, 2024 [11]. Later 2024 work connected the program to Lie-group transformations, Homomorphism Autoencoders, sheaf transport, tomography, variability, and neural diffusion [12,13]. Some 2024 source records are mixed human-AI dialogues. They can establish the chronology of Micah-attributed questions and corrections, but generated replies remain discovery and editorial material rather than independent scientific evidence. The present paper adds the executable model-selection and process-equivalence benchmark in 2026. No mature term is backdated to an earlier precursor.

Stage	Recoverable atom	Date boundary	What the source does not establish
2012 notes	Graph, vector, category, topology, and differential-equation research questions	Author-dated 2012; public Git bytes in 2022	A completed neural formalism
2014 note	Learned active structure as EDS ancestor	Author-dated 2014; public Git bytes in 2022	A standard software object inside a dendrite
2017 podcast	Feedback/no-feedback communication problem and ATP analogy	Public recording in 2017	Literal TCP or UDP in tissue
June-July 2022	Nested place-process ledger; ordered phase/Taylor trajectory; inhibition and path selection	Immutable public Git commits	Biological validation of a categorical or Taylor implementation
2024	Named NDCA; transport, symmetry, tomography, variability, and diffusion synthesis	Public Git plus bounded mixed dialogues	One completed cross-scale decoder
2026	Typed cross-scale model and deterministic selection benchmark	Current preprint	Evidence that neural tissue implements the benchmark

3. The inherited mathematical foundation

Neural state-space analysis represents population activity as trajectories rather than isolated labels. Movement-related cortical dynamics and context-dependent prefrontal computation demonstrate why a population trajectory can reveal lawful structure that is obscured at the level of one unit [15,16]. Dimensionality reduction can make such structure analyzable, but the fitted axes remain analyst-defined unless a downstream biological receiver is shown to use the relation [17]. Representational geometry likewise compares patterns through declared dissimilarities and transformations; it does not guarantee one native coordinate system across brain regions [18].

History dependence creates a second inherited foundation. A current sample may be insufficient when the output depends on earlier inputs or states. Delay embeddings, state-space models, Volterra kernels, recurrent models, and nonlinear system-identification methods offer different ways to represent that history [19-22]. A Taylor expansion is appropriate only locally, under smoothness and controlled remainder. It cannot be privileged merely because the SAN genealogy contains a Taylor analogy.

Local coordinate systems create a third foundation. Cellular sheaves assign local vector spaces and restriction maps to nodes and edges; Neural Sheaf Diffusion and Sheaf Attention Networks demonstrate that learned transport can change the behavior of graph models [23,24]. These engineering results make transport a coherent comparator. They do not establish that cortical columns are literal sheaf modules.

Compositional process theory creates a fourth foundation. Open Petri nets expose input and output places so processes can be joined; categorical treatments distinguish execution semantics and preserve composition; open reaction networks can be mapped to open dynamical systems [25-27]. These tools make it possible to record where an event happens and how processes connect without equating endpoint agreement with isomorphism.

Structured transformations and local learning supply further comparators. Homomorphism Autoencoders test whether action composition is preserved in a learned representation [28]. Burst-dependent and compartment-specific learning models show how locally available dendritic and feedback signals can coordinate a defined hierarchical learning task [29]. Nonlinear dendritic integration demonstrates that the receiver's branch and state can alter the transformation applied to nearby inputs [30]. The SAN program inherits these mathematical and biological facts, then asks whether a particular joined architecture adds held-out and causal value.

4. Typed state, receiver, and event

4.1 A trajectory requires declared measurements

Let a measured system at scale s have state $x_s(t)$, input $u_s(t)$, parameters θ_s , and omitted influence $\eta_s(t)$:

$$\frac{dx_s}{dt} = F_s(x_s, u_s; \theta_s) + \eta_s. \quad (1)$$

Equation (1) is an analyst's model. Every application must declare coordinates, units, sampling window, observation map, initial and boundary conditions, and the intervention under which trajectories are compared. A low-dimensional trajectory is not automatically a semantic code, attractor, or conscious rendering.

4.2 The same input can meet a different receiver

Let the receiver state be a typed bundle

$$E_r(t) = \{m_r, s_r, g_r, i_r, h_r, c_r\}, \quad (2)$$

where the fields may represent morphology, synaptic configuration, conductance state, inhibition, recent history, and context. The response is

$$y_r(t) = F_r(u_r(t), E_r(t)). \quad (3)$$

The braces in Equation (2) do not authorize arithmetic among heterogeneous fields. Numerical use requires a declared encoding map

$$\psi_r: E_r \rightarrow R_r^d. \quad (4)$$

Only now is EDS earned as a compact name for the proposed learned receive-transform-project configuration. The hypothesis is removed where the declared bundle adds no held-out or causal value beyond a point-neuron, compartmental, active-dendrite, or recurrent-state model.

4.3 A consequential difference is receiver-relative

One arrival can be represented as a typed event record

$$p_{i \rightarrow r}(t) = [\Delta\phi, \Delta f, \Delta A, \Delta\tau, \Delta q, \Delta v, \Delta x]_{i \rightarrow r, t}. \quad (5)$$

The fields can describe phase relation, event rate or frequency, amplitude, duration, transmitted-quantity proxy, variability, and spatial or network state. They occupy a product space with different units. A numerical feature vector exists only after a frozen measurement contract supplies scaling and encoding:

$$d_{i \rightarrow r}(t) = \chi_{i \rightarrow r}(p_{i \rightarrow r}(t)). \quad (6)$$

This is the earned mathematical role of PWD in this paper: a preregistered, typed, receiver-relative difference associated with an identified sender, route, delay, receiver, reference, and endpoint. A raw phase

difference lacks several of those fields. A traveling wave additionally requires spatial propagation. A prediction error can overlap with one component without exhausting the event family.

5. Order, local approximation, and history

A state at one moment cannot recover an ordered transformation by itself. Near an operating point x_0 , a sufficiently smooth receiver map can be approximated by

$$P_K(\Delta x) = \sum_{k=0}^K \frac{1}{k!} D^k F_r(x_0) [\Delta x]^{\square k}. \quad (7)$$

The corresponding local error is

$$R_{K+1}(\Delta x) = F_r(x_0 + \Delta x) - P_K(\Delta x). \quad (8)$$

A Taylor-specific account is warranted only inside a declared neighborhood N where

$$\sup_{\Delta x \square N} \|R_{K+1}(\Delta x)\| \leq \varepsilon_R. \quad (9)$$

Thresholds, switching, bifurcations, noise, and long memory can violate that local description. A general finite-history comparator can instead be written

$$y_r(t) = G_r(u(t), u(t-1), \dots, u(t-L), E_r(t)). \quad (10)$$

Volterra, recurrent, spline, autoregressive, and switching models instantiate different versions of G_r . The scientific question is not whether Taylor or recurrence sounds more compatible with SAN. It is which minimum model predicts held-out consequences and survives a selective perturbation in the declared regime.

6. Local coordinates, delay-aware transport, and process composition

6.1 Different local views need maps rather than forced sameness

Suppose local populations u and v encode related variables in spaces X_u and X_v . Their mapped agreement on edge e is measured by

$$r_e(t) = T_{u \rightarrow e} x_u(t - \tau_{u,e}) - T_{v \rightarrow e} x_v(t - \tau_{v,e}). \quad (11)$$

Compatibility is a claim only relative to a norm, tolerance, units, and aligned times:

$$\|r_e(t)\|_e \leq \varepsilon_e. \quad (12)$$

The local states need not be identical. The maps T are estimated from anatomy, calibration, or training data and then frozen before evaluation. If delays are omitted, Equation (11) can compare different biological moments while appearing algebraically valid.

6.2 A transformation must have compatible interfaces

Let an open process $P_a: X \rightarrow Y$ produce a typed output consumed by $P_b: Y \rightarrow Z$. Their sequential composition is

$$P_b \square P_a : X \rightarrow Z. \quad (13)$$

Parallel composition of independent declared interfaces is

$$P_a \square P_c : X \times W \rightarrow Y \times V. \quad (14)$$

Equations (13) and (14) are a process ledger, not a claim that biological signals literally manipulate categorical symbols. They require matching domains and codomains, declared resources, timing, and a measurement at intermediate places. A chemical and an electrical route may agree at one output while differing in intermediate state, reversibility, energy use, side effects, or future reachability.

6.3 Identity and action transform differently

For a declared transformation g , approximate invariance and equivariance are

$$\begin{aligned} D(z(gx), z(x)) &\leq \varepsilon_I \\ z(gx) &\approx \rho(g)z(x). \end{aligned} \quad (15)$$

A group-structured claim additionally requires composition consistency:

$$\|\rho(g_1 g_2) - \rho(g_1)\rho(g_2)\| \leq \varepsilon_G. \quad (16)$$

Object identity may remain stable while position and orientation transform for action. A successful equivariant decoder does not by itself establish a Lie group in neural tissue; the identity, transformations, held-out compositions, and behavioral consequences must be specified.

7. Partial views and the cross-scale joining problem

Let a hidden state $z(t)$ generate partial observations

$$o_j(t) = H_j z(t) + v_j(t), \quad j = 1, \dots, m. \quad (17)$$

A reconstruction model estimates

$$z(t) = R(o_1(t), \dots, o_m(t), H_{<}). \quad (18)$$

The reconstruction claim is evaluated with a declared loss and downstream endpoint:

$$\begin{aligned} L_{rec} = \\ E \|z - \hat{z}\|^2 \\ + \lambda L_{action}. \end{aligned} \quad (19)$$

These equations state the problem that earns the NAPOT name. Genuine tomography requires a forward model, sampling geometry, noise model, observability or identifiability conditions, and reconstruction error. External decodability alone does not show that a biological receiver performs or uses the reconstruction.

The broader NDCA problem joins bounded scale-specific models. Let each scale have a readout $y_s = R_s(x_s)$. A delay-aware cross-scale state is

$$x_{s+I}(t) = C_s \left(T_{l \rightarrow s} y_{s,l}(t-\tau_l), \dots, T_{n \rightarrow s} y_{s,n}(t-\tau_n) \right). \quad (20)$$

Equation (20) does not assert that every scale is smooth or equivalent. It defines the minimum information needed to test a cross-scale composition: states, dynamics, readouts, transport maps, delays, approximation target, error, receiver, and rejection rule.

8. Four formal propositions that constrain interpretation

Proposition 1: receiver dependence defeats a source-only deterministic map

If there exist receiver states $e_1 \neq e_2$ and an input u such that $F(u, e_1) \neq F(u, e_2)$, then no single deterministic source-only function $g(u)$ represents both outcomes for that input.

Proof. A function assigns one output to one input. If $g(u) = F(u, e_1)$, then it cannot simultaneously equal the distinct value $F(u, e_2)$. The data can still be approximated by averaging or noise, but the receiver variable is structurally omitted. This proposition does not prove which biological receiver variables are sufficient.

Proposition 2: typed composition requires interface and time alignment

The composition $P_b \square P_a$ is defined only when the output type of P_a matches the input type of P_b , including declared units, encoding, and temporal support. Algebraic dimension agreement without time alignment can compose measurements from different events.

Proof. Composition requires that every output element of P_a belong to the domain on which P_b is defined. A value measured at $t-\tau_a$ is not the same typed event as a value at $t-\tau_b$ unless a transport or alignment rule identifies them. The mismatch can be hidden by equal array dimensions, so dimensionality alone is insufficient.

Proposition 3: endpoint equality does not establish process isomorphism

Let O be an observation map and P_A, P_B two processes. If

$$O(P_A(x)) = O(P_B(x)) \text{ for all tested } x, \quad (21)$$

then the routes are observationally equivalent relative to O on that test. Equation (21) does not imply $P_A = P_B$, an invertible map between their internal states, or preservation of composition.

Proof. A many-to-one observation map can send distinct internal trajectories to one endpoint. An isomorphism requires a structure-preserving invertible relation, while Equation (21) constrains only the observed image. Intermediate measurements or interventions can distinguish the routes.

Proposition 4: latent coordinates have a gauge ambiguity

For an invertible matrix G , the observation model $o = Hz$ is unchanged under

$$z'=Gz, \quad H'=HG^{-1}, \quad H'z'=Hz. \quad (22)$$

Therefore latent coordinates and transport maps are not uniquely identified by reconstruction error alone. Anatomical anchors, interventions, known transformations, or additional constraints are required before a fitted coordinate is promoted to a biological variable.

9. Deterministic cross-scale benchmark

9.1 Why a new benchmark was necessary

The existing PWD calculus paper already compares Taylor, Volterra, autoregressive, and typed-PWD models and reports adverse approximation and reconstruction results. Repeating that synthetic program would create redundant publication. The present benchmark owns a different question: can one selection procedure recover when receiver state, history, coordinate transport, and process composition are separately required, and can it refuse those structures when they are unnecessary?

9.2 Data generator and fixed split

Each synthetic sequence contains a two-dimensional latent state with stable recurrent dynamics, two noisy local views, a receiver context, a coordinate rotation, and a scalar consequence. The first view is expressed in the reference coordinates. The second is rotated by a sequence-specific angle. Train, validation, and test sets contain separate complete sequences. Test sequences use rotations not present in training. No seed is removed.

Four regimes add one generating dependency at a time:

- Instantaneous: the consequence depends only on the current reference view.
- Receiver: the same current view has a context-dependent consequence.
- History: receiver dependence remains and lagged reference states add predictive structure.
- Transport composition: a delayed second view must be inverse-transported into the reference coordinates and composed with the current state.

The full synthetic target has the form

$$\begin{aligned} y_t = & \beta^T a_t \\ & + q \gamma^T a_t \\ & + \sum_{\ell=1}^2 \delta_{t-\ell}^T a_{t-\ell} \\ & + \kappa^T T(-\theta) b_{t-2} \\ & + a_t^T M T(-\theta) b_{t-2} \\ & + \varepsilon_t. \end{aligned} \quad (23)$$

Terms are activated cumulatively by regime. The generator is intentionally transparent. It tests whether the pipeline can recover known structure, not whether neural data contain that structure.

9.3 Candidate models and selection rule

Four models receive successively richer information:

Model	Available information	What it tests
M0 instantaneous	Current reference view	Is present state sufficient?
M1 receiver	M0 plus context and input-context interactions	Does the same input differ by receiver?
M2 history	M1 plus two reference lags and untransported second-view history	Is temporal history needed without an explicit coordinate map?
M3 transported composition	M2 plus delay-aligned inverse transport and cross-view composition terms	Do coordinate transport and composition improve held-out generalization?

Every model is fitted only on training sequences. The smallest model whose validation RMSE lies within three percent of the best validation RMSE is selected:

$$m = \min\{m: RMSE_{val}(m) \leq 1.03 \min_j RMSE_{val}(j)\}. \quad (24)$$

The selected model is then evaluated once on held-out sequences. The tolerance gives the simpler model priority when extra structure produces only a negligible improvement. Thirty fixed seeds, 0 through 29, are included.

9.4 Route-equivalence test

A separate generator constructs routes A and B with the same source and nearly identical normal endpoint distribution. Their intermediate states differ and a blocked-transition intervention exposes that difference. A nearest-centroid classifier is trained either on source-plus-endpoint features or on the full typed path. The purpose is not to demonstrate a sophisticated classifier. It is to show, constructively, what Proposition 3 says: an endpoint observation can erase the information needed to distinguish processes.

10. Results

10.1 The minimum adequate model was recovered in every regime

The selection rule recovered the expected model in 30 of 30 runs for each of the four regimes. The full 120-run recovery result is therefore 120 of 120. This perfect result reflects a deliberately identifiable synthetic generator and should not be read as an estimate of biological model-selection accuracy.

Regime	Expected selected model	Recovery	Median held-out RMSE M0	M1	M2	M3
Instantaneous	M0	30/30	0.129	0.130	0.130	0.130
Receiver	M1	30/30	0.495	0.130	0.131	0.131
History	M2	30/30	0.638	0.448	0.132	0.132
Transport composition	M3	30/30	1.178	1.080	0.319	0.131

The most important control is the first row. M3 was available in the instantaneous regime but was never selected. The procedure therefore did not reward complexity merely because transport and composition were present in the vocabulary. Receiver variables produced a large gain only when the generator made receiver context consequential. History produced a further gain only in the history regimes. In the cross-scale regime, untransported history retained substantial predictive value, but explicit delay-aligned inverse transport and composition reduced median test RMSE from 0.319 to 0.131.

Transport itself passed a numerical invariance check. Across 5,000 random two-dimensional states and rotations, inverse transport reconstructed the original state with maximum Euclidean error 6.28×10^{-16} . This verifies the implementation of the coordinate map, not the existence of a neural transport operator.

Cross-scale benchmark: median held-out RMSE

Lower is better; bold border marks the expected minimum adequate model

	M0 instantaneous	M1 receiver	M2 history	M3 transported composition
Instantaneous	0.129 recovery 100%	0.130	0.130	0.130
Receiver	0.495	0.130 recovery 100%	0.131	0.131
History	0.638	0.448	0.132 recovery 100%	0.132
Transport Composition	1.178	1.080	0.319	0.131 recovery 100%

Design rule

Select the smallest model within 3% of the best validation RMSE; evaluate once on held-out sequences.

The transported model is not rewarded in regimes where simpler structure is sufficient.

Figure 1. Minimum adequate model recovery. Median held-out RMSE across 30 fixed seeds is shown for every model in every generating regime. The bold diagonal identifies the expected minimum adequate model. The selection rule recovered that model in 30 of 30 runs per regime while refusing M3 when simpler structure was sufficient.

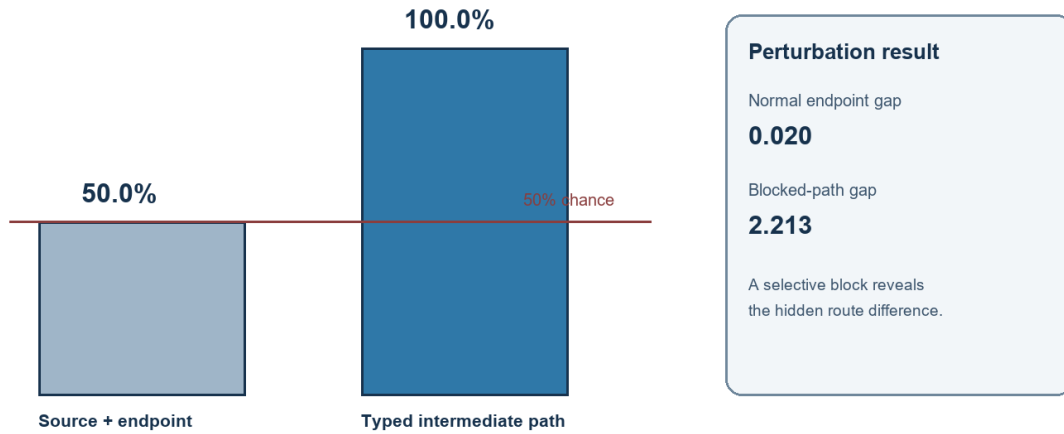
10.2 Matching endpoints concealed the route

The route-equivalence benchmark used 8,000 samples. A classifier receiving source and endpoint achieved 50.04% held-out accuracy, effectively chance. Adding the typed intermediate states raised accuracy to 100%. The mean normal endpoint difference between routes was 0.020. Blocking the compensating intermediate transition exposed a mean difference of 2.213.

Observation	Route accuracy or separation	Interpretation
Source plus normal endpoint	50.04% accuracy	Endpoint map erases route identity
Source plus typed intermediate path	100% accuracy	Intermediate process record preserves route identity
Mean normal endpoint gap	0.020	Routes are observationally matched at the declared endpoint
Mean gap after intermediate block	2.213	Selective perturbation reveals the hidden route difference

Equal endpoints do not establish equal mechanisms

Route discrimination before and after exposing the typed process path



Endpoint agreement supports observational equivalence only relative to the declared observation map and tolerance.

Figure 2. Endpoint equivalence versus path identity. Source-plus-endpoint measurements remain at chance for route classification, while typed intermediate measurements recover route identity. Blocking the compensating transition reveals a large route-specific separation that the normal endpoint hides.

The result supplies an executable distinction among equality of endpoints, observational equivalence, mechanistic identity, and isomorphism. It also explains why a category-Petri process ledger is useful even when a state-space model predicts the final label: intermediate states and interventions can determine which process actually occurred.

10.3 Claim ceiling

The benchmark is favorable because the synthetic regimes were constructed to contain the named dependencies. It validates data splitting, typing, transport, model selection, null preference for simpler structure, path discrimination, and intervention logic. It does not establish that EDS, PWD, NAPOT, or NDCA is implemented in neural tissue. It does not show that known coordinate maps are available in biological experiments. It does not make synthetic perfect recovery a probability that the theory is true.

11. How the nineteen mathematical families fit without becoming one equation

Family	Role in the modular program	Strong control or alternative	Failure removes
Directed graph and recurrent paths	Typed routes, loops, delays, and feedback	Anatomy, effective connectivity, functional association	The claimed graph mechanism
Category theory and Petri nets	Interface-compatible sequential and parallel processes	State-space, compartmental, reaction-network, causal graph	Category-Petri-specific composition
Vector and representational geometry	Declared multivariable coordinates	Raw measurements and alternate coordinate systems	The selected representation
Differential equations	Continuous or hybrid state change	Switching, stochastic, and event models	The proposed dynamics
Phase space and attractors	History-dependent trajectories and regimes	Rate and recurrent alternatives	The claimed trajectory coordinates
EDS	Learned receiver state bundle	Point, compartmental, and active-dendrite models	Unsupported EDS fields
ATP ancestry	Communication-role question	Biological signaling and protocol controls	Literal network-protocol analogy

Family	Role in the modular program	Strong control or alternative	Failure removes
PWD	Typed receiver-relative event difference	Rate, phase, waveform, prediction error, recurrent state	Unsupported PWD dimensions
Taylor approximation	Local value, slope, curvature, and interaction terms	Splines, Volterra, recurrence, switching	Taylor-specific implementation
History models	Dependence on earlier input and state	Capacity-matched alternative memory models	Claimed memory family
Reaction-diffusion and bifurcation	Spread, stabilization, extinction, and route change	Accumulator, control, adaptation, basal ganglia	Claimed route operator
Invariance and equivariance	Stable identity and predictable transformation	Augmented decoder and generic latent dynamics	Group-structured claim
Lie groups and HAE comparison	Composition of actions	Useful equivariance without a group	Homomorphism claim
Sheaf compatibility	Transport among local views	Ordinary graph and manifold alignment	Sheaf-specific transport
Tomography	Reconstruction from partial views	External decoder without endogenous use	Tomography claim
Neural Diffusion Network	Redistribution into later readiness	Adaptation, inhibition, homeostasis, recurrence	SAN diffusion hypothesis
Shannon information and variability	Ensembles, surprisal, conditional target value	Log-loss and conventional statistics	Unsupported information claim
Dendritic credit assignment	Locally available learning signals	Burst, feedback, eligibility, plasticity models	Universal analog-backprop claim
NDCA	Cross-scale joining and approximation test	Independent scale-specific models	NDCA as a unified formalism

This modular interpretation is scientifically stronger than an all-or-nothing theory. Failure of a sheaf-specific map need not erase receiver dependence. Failure of Taylor approximation need not erase history dependence. Failure of tomographic reconstruction need not erase ordinary recurrent integration. Conversely, success of one component does not validate the rest.

12. Translation to a biological experiment

The synthetic benchmark identifies the structure of a decisive biological study. Record from identified source and receiver populations while also measuring behavior, movement, arousal, waveform features, and the receiver state relevant to the preparation. Use complete trials as the split unit so adjacent samples and paired receivers do not leak across training and evaluation. Fix filters, phase estimators, coordinate maps, delays, feature scaling, and outcome definitions without evaluation labels.

The analysis should compare, at minimum:

1. current source state;
2. source plus receiver state;
3. source, receiver, and history;
4. an untransported multi-area model;
5. a delay-aware transported composition model;
6. a capacity-matched recurrent state-space model;
7. preparation-specific predictive-processing, control, or communication-subspace alternatives.

The strongest intervention does not simply silence a region. It changes one predicted interface relation while preserving total input, gross firing, movement, and arousal as closely as possible. If the model claims a delay-aware route, shift timing without changing mean input. If it claims a transport map, disrupt or remap the relation between local coordinates. If it claims a process composition, block one intermediate transition. A

rescue should restore the specific relation rather than generic excitability.

The model earns cross-scale explanatory value only when it improves held-out prediction on unseen conditions, identifies a receiver-specific consequence, predicts the selective perturbation, and supports a matched rescue. Similar-looking trajectories or successful external decoding are not sufficient.

13. Nonduplication and relationship to companion papers

The PWD methods paper owns the minimal operational estimator and its biological-development benchmark. The PWD calculus paper owns the broader typed-event calculus, Taylor/Volterra comparison, synthetic tuning fixtures, and their adverse results. The NAPOT methods paper owns the frozen reconstruction benchmark and its source genealogy. The Choice paper owns inhibitory route selection and commitment. The EDS and Artificial Neurology work owns the broader learned-structure and engineering implications.

This paper contributes a different object: a delay-aware typed composition framework plus a model-selection benchmark that can choose among instantaneous, receiver, history, and transported-composition structures. It also contributes the endpoint-versus-path discrimination experiment and the four formal propositions. Shared definitions are included only to make the cross-scale model intelligible. The adverse PWD results are not converted into favorable evidence here, and no benchmark is presented twice as an independent confirmation.

14. Falsifiers and narrowing rules

The framework must narrow rather than expand after failure.

- If receiver variables do not improve held-out prediction or selective intervention, the receiver-specific extension is removed for that preparation.
- If history accounts for the data without a PWD or transport term, the history result remains and the unsupported term is retired.
- If untransported recurrent models generalize as well as the declared coordinate map, the transport-specific claim fails.
- If endpoint and intermediate measurements remain indistinguishable under targeted perturbation, the process-ledger claim has not added evidence.
- If learned coordinate maps change across folds, references, or sessions without a stable anchor, they remain analyst conveniences.
- If a reconstruction is externally decodable but no downstream receiver uses it, the endogenous tomography claim fails.
- If a local model works but cross-scale composition adds no held-out or causal value, NDCA remains a collection of scale-specific models rather than a unified formalism.
- If the biological variables cannot be measured without using the outcome to define them, the test is circular.

Failure of one proposed correspondence does not turn the original source wording into an error. It identifies which mathematical interpretation or biological extension did not survive the declared test.

15. Limitations

The synthetic generator is low dimensional, stable, and deliberately identifiable. Its coordinate rotations are exactly known. Its noise is mild. Its receiver variable is observed. Real neural data contain missing variables, drifting representations, correlated noise, movement, shared input, estimator uncertainty, nonstationarity, and ambiguous scale boundaries. A perfect synthetic selection rate is therefore a software and logic result, not an empirical confidence level.

The propositions are formal constraints on model interpretation, not discoveries about neural tissue. Proposition 1 says when a source-only deterministic map is structurally insufficient; it does not identify the correct receiver state. Proposition 2 requires typed and temporal interfaces; it does not prove biological category structure. Proposition 3 distinguishes endpoint equivalence from process identity; it does not accuse any biological model of concealing a route. Proposition 4 identifies latent-coordinate ambiguity; it does not make every representation arbitrary.

The nineteen families also differ in maturity. State-space analysis, Shannon information, differential equations, Petri nets, sheaves, group actions, and system identification are established mathematics. EDS, PWD, NAPOT, Neural Diffusion Network, and NDCA are SAN proposals whose scientific status depends on the stated tests. One useful mapping cannot validate a complete consciousness theory.

16. Conclusion

The mathematical problem begins with a simple biological fact: the same event can meet a different receiver and enter a different history. From there, a serious model must preserve types, delays, local coordinates, process interfaces, and observable consequences. State trajectories describe change. Receiver bundles describe learned local transformation. Typed differences describe candidate consequential departures. Taylor and history models describe ordered change under different assumptions. Transport maps compare local views. Open processes record how events compose. Invariance, equivariance, and tomography ask what identity survives and what a collection of partial views can reconstruct. NDCA names the remaining work of joining those pieces across scale without pretending they are one universal equation.

The new benchmark turns that program into an executable selection problem. It chooses the simplest adequate model when current state is enough, adds receiver state only when the same input has receiver-dependent consequences, adds history only when earlier states matter, and adds coordinate transport and composition only when they improve generalization to unseen transformations. It also demonstrates why equal endpoints cannot establish equal mechanisms: a typed intermediate record and a selective perturbation can reveal distinctions that the endpoint map removes.

The result is not a proof that the brain implements SAN mathematics. It is a stronger starting point for testing that possibility. The next step is a preregistered biological study in which each mathematical addition must earn its place through held-out prediction, route-specific perturbation, and matched rescue.

Data and code availability

The deterministic benchmark script, fixed seeds, complete run table, machine-readable results, validation receipt, and generated figures are included in the companion release package. The script uses only NumPy and Pillow in addition to the Python standard library. No human or animal data were analyzed. Historical private-source paths are preserved only in the governed source manifest and are not released as data.

Authorial provenance

Micah Blumberg is the sole author. The paper develops the Self Aware Networks mathematics program from dated first-party sources and adds the 2026 formal propositions, benchmark design, implementation, interpretation, and falsifiers. AI systems assisted with source routing, code execution, consistency checks, formatting, and adversarial review under the author's direction. AI output is not treated as an independent scientific source or co-author.

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